

Sample-efficient inverse focal-point synthesis for HIFU phased arrays via gauge-aware reformulation

Background. Steering a 256-element HIFU phased array onto a desired focus in heterogeneous tissue is conventionally posed as inverse phase regression, but the map is doubly ill-posed: empirically $Q \rightarrow \text{phase}$ is one-to-many (nearest-neighbour phase-vector circular-cosine similarity 0.002 ± 0.043 , indistinguishable from noise) and the system carries a continuous gauge symmetry — adding a constant $+\delta$ to every phase produces zero focal displacement and sub-1 % intensity change. Direct deterministic regression cannot fit this combination.

Method. We reformulate the task as **3-DOF focus-point regression**: a compact 3-D CNN (FocusPointNet, 0.89 M params) predicts (x, y, z) from $(\log_{10}(Q), \text{mask})$; transducer phases are then synthesised by **analytical delay-and-sum** beamforming, gauge-fixed to $\phi_0 = 0$ and quantised to 5° steps. The gauge symmetry was verified three independent ways — analytic derivation from the delay-and-sum equation, synthetic 256-element least-squares fit, and full-wave **k-Wave simulation** (ITU collaborator) — all reporting **0 mm focal shift and <1 % intensity change at $+20^\circ$** .

Results. On a 22 / 4 / 4 split with three random seeds, FocusPointNet achieves **lateral RMS 4.63 ± 2.21 mm (X) and 2.69 ± 1.21 mm (Y)** — well inside the 6 mm focal spot — combined RMS **26.25 ± 4.61 mm**, dominated by the elongated focal zone along the axial axis. Backbone ablation (CNN / ResNet-3D / multi-scale UNet-encoder) and a state-of-the-art DSNT heatmap variant (nnLandmark / H3DE-Net 2025) reach equivalent lateral accuracy: **precision is a property of the task formulation, not of any specific backbone**. Against five closed-form classical baselines on the same split, FocusPointNet improves lateral X by **$3.0\times$** and Y by **$5.3\times$** over the strongest (amplitude-weighted centroid: 13.82 / 14.23 mm); simpler peak-finders collapse to 64–80 mm RMS because the heat peak is offset from the intended target by 30–50 mm under refraction. A $5^\circ / 10^\circ / 15^\circ$ quantisation study confirms <0.35 % intensity error and sub-0.1 mm focal shift at 5° , enabling a discrete gauge-fixed output space.

Significance. To our knowledge this is the first **gauge-aware reformulation** of HIFU phased-array inverse synthesis as a low-dimensional regression task, recovering single-valued ground truth while preserving the underlying physical symmetry. The empirical gauge-and-quantisation characterisation is of independent value for classical phase optimisers, and the sample-efficient learned-localisation + analytical-beamforming hybrid achieves sub-focal-spot lateral accuracy from only 22 training simulations.

Keywords: HIFU, phased array, gauge symmetry, focal-point regression, delay-and-sum, phase quantisation, sample efficiency.